

**ADMINISTRATIVE INFORMATION****FMI No: 25406****FMI Title:** VCT External Evaluation Sites Inspection and upgrade

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

- ☒ **Staged FMI** **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
- ☐ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
- ☐ **Parts Ordering:** GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator
- ☐ **No Parts** are required to complete this FMI.
- ☐ **Batch with** _____
To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
- ☐ **Non-Staged** **No Customer List is provided.** Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Online Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

The purpose of this FMI is to inspect the original VCT External Evaluation sites listed in the effectivity for proper manufacturing configuration and upgrade where needed.

Any questions, please contact Kevin Swain, MI&CT Lead Service Integrator at:

Phone #: 262-312-7046 or

Cell #: 262-424-5281.

Unit Tracked:**FMI Instructions Part No.: 5270129-1EN****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
GE Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

FMI No.: 25406

FMI Title: VCT External Evaluation Sites Inspection and upgrade

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Revision History

Revision	Date	Reason for Change
1	12/07/2007	Initial Release

Effectivity

This FMI applies **ONLY** to the following VCT External Evaluation Sites. No other serial numbers should be added to the effectivity.

System ID	Serial Number	Gantry Model	Hospital Name
414805CT4	359252CN4	5124069	Froedtert Memorial (GEMSAM)
732321VCT	359394CN4	5124069	JFK (GEMSAM)
M2322223	359588CN1	5124069	CCN (GEMSE)
XG0009	361977CN2	5124069	Keio (GEMSA)
828210048	364830CN0	5124069	Hong Kong Sanatorium (GEMSA)

Purpose

The purpose of this FMI is to inspect the original VCT External Evaluation sites listed in the effectivity for proper manufacturing configuration and upgrade where needed. The areas that need to be checked are:

- Hardware configuration
- Installation of parts to upgrade engineering components
- Existence of all Labels
- Record of Safety and Regulatory tests

Prerequisite FMIs

None

Furnished Materials

FMI 25406, Kit Part Number 5270129		
Qty	Description	Part #
1	FMI 25406 Document	5270129-1EN

Personnel Requirements

Travel Time: 1 hour for each trip needed to complete the FMI

Labor Time: Labor time depends upon any upgrade required to bring the system to the production equivalent.

Procedure

Before you begin:

Before starting the hardware inspections, power down the system **and follow Lockout-Tagout per the following procedure:**

Follow these general rules:

- All installed GE Healthcare systems require a LOTO compatible A1 Disconnect.
- Only qualified service personnel trained in the service and operation of this scanner, options, and accessories should perform any service on this equipment. Service personnel must observe and apply LOTO as defined by annual LOTO training.
- Equipment fuses, switches, and circuit breakers are for fire and equipment protection only. Do not rely on them to protect you against electrical shock or un-commanded equipment motion. This includes the Gantry Service Switch Panel!

Remove Power from the System

1. Prepare for shutdown of equipment. Notify affected personnel working in the area, via verbal communication, that LOTO is being performed.
2. Bring the system software down at the operator console.
3. After software shutdown is complete, wait two minutes and then place console power switch in the OFF position.
4. Turn off power at the system A1 Panel or Main Disconnect Panel. All personnel servicing the system must place their personal LOTO devices and tags on the A1 Panel or Main Disconnect Panel.
5. Verify that all energy has been dissipated as follows. Take the top and front cover off the PDU. Measure incoming system power at TS1 (for NGPDU) or Input Power Panel (for Compact PDU). Measure each phase to ground and verify zero potential.
6. Wait 5 minutes for any stored energy in capacitors to dissipate.
7. Service/repair the CT system per service instructions.

Restore Power to the System

1. Notify affected personnel that equipment will be re-energized and LOTO devices are being removed.
2. All service personnel are to remove their personal LOTO lock and tag from A1 Panel or Main Disconnect Panel.
3. Re-apply power to the PDU Cabinet.
4. Reset the Gantry drives by pushing the reset button on any of the control panels.
5. Bring the system software back up.

The FMI will not be closed until:

- ***Inspection is completed:***
- ***Upgrade tasks, if needed, are completed and***
- ***Completed checklist is returned to Kevin Swain, MI&CT Lead Service Integrator (step 6).***

The completion of this FMI may take multiple trips to the site if upgrade is needed.

1. Hardware Configuration:

Check hardware model numbers of Table, Collimator, Detector and Aux Box. The expected part numbers are listed in the Checklist attached. If any of the components is not the expected part, contact Headquarters listed in step 6 for required action.

2. Existence of all Labels

Check to make sure all required labels listed in the Checklist exist. For missing labels, contact Headquarters listed in step 6.

3. Installation of parts to upgrade engineering components

Parts and installation instructions were sent to all the External Evaluation sites before the product release to update certain components for serviceability and to the manufacturing level. Please inspect to see whether this activity was completed (Appendix B). If any of the items are not completed, contact Headquarters listed in step 6 for required action.

4. Safety and Regulatory Tests

KV/Ma Verification and Exposure Time Accuracy tests should have been done and the results documented. If these were not done, perform these tests (see [LightSpeed 7.X Limited Access Service Methods - Advanced manual, cdrom # 5193575-800](#)) for instructions as to how to perform these tests). This does not apply to all the sites. Please see the Checklist for the sites requiring inspection. Once the tests are completed successfully, send the results to Headquarters listed in step 6.

5. Software:

All sites should be running either 06MW3.4, 06MW29.7 or 07MW18.4 (FMI 25394) software.

Please inspect to make sure that the site is running one of the above software and complete the checklist. If the site is not running the correct software, contact Headquarters listed in step 6 for required action.

6. Headquarters contact

Complete the Checklist attached with this FMI for the inspection as indicated in steps 1 through 5. If any of the checks need action, please contact Kevin Swain for required action. He can be contacted at:

Kevin Swain

MI&CT Lead Service Integrator

Phone #: 262-312-7046

Cell # 262-424-5281

Email kevin.swain@med.ge.com

Send the completed Checklist to Kevin Swain.

7 Check lists

System ID: _____

Field Engineer: _____

Date: _____

7.1 Hardware Configuration:

	Component	Expected Part	Actual Part #	Write down serial number	If part # is different than Expected Part, contact Headquarters and write down resolution method and date
1	Table	5121647			
2	Collimator	5130001			
3	Detector	5126400			
4	Aux Box	2324118-3			

7.2 Labels check: (See Appendix A)

	Component	Label	Figure #	Correct label exist? Yes/No	If No, contact Headquarters and write down resolution method and date
1	Gantry	G1	1		
		G2	1		
		G3	1		
		G4	1		
2	Aux Box	A1	2		
3	Table	T1	3		
		T2	3		
		T3	3		
4	Console	C1	4		
		C2	4		
		C4	4		
		C5	4		
		C6	4		
5	PDU	P1	5		
		P2	5		
		P3	5		
		P4	6		
		P5	6		
		P6	6		

7.3 Installation of Hardware Components (See Appendix B)

Part Number	Description	Installed? Yes/No	If No, contact Headquarters and write down resolution method and date
2401577	SCAN WIN ASSY - VCT GANTRY		
2397075	COVER ASSY, FRONT GANTRY-VCT		
2397078	COVER ASSY, REAR GANTRY-VCT		
5130600	Harness Top Cover DC Fans		
5129522	Cable - Table to TGPU - J28		
5125258	Fiber Optic - Receivers to Power Pan		
5125259	Fiber Optic - Console to Gantry		
5126690	keeper,slam latch, VCT, symmetric		
5118424	Harness - Filter to GFI		
5118296	Harness - GFI to Brush block		
5125899	service board assy VCT		
5125984	cover, circuit breaker		
5125905	slip ring plate assy_VCT		
5130598	Harness Tilt Side Fan Control		
5126690	keeper,slam latch, VCT, symmetric		
5128605	bulkhead bracket,48v PS,VCT		
5128606	guard,48v PS,VCT		
5122691	VCT front cover latch assembly		
5119936	cover, terminal strip		
5124826	safety panel-ls rear-VCT		
5124827	bracket,ls rear safety panel_VCT		
5124826	safety panel-ls rear-VCT		

7.4 Safety and Regulatory Tests (Only required for the sites listed):

	Site ID	Test	Test performed with successful results? Yes/No or Not Applicable	If No, contact Headquarters and write down resolution method and date
1	M2322223	KV/Ma Test		
	M2322223	Exposure Time Accuracy		
2	732321VCT	Exposure Time Accuracy		
3	828210048	Exposure Time Accuracy		

7.5 Software Version Test:

	Expected Software	Actual Software	If different than the three software versions, contact Headquarters, and write down resolution method and date
	06MW3.4, 06MW29.7 or 07MW18.4		

8. FMI Closure

The FMI can be closed only after:

- All the required actions are completed and***
- The check list completed and returned to Headquarters.***

Once all actions are complete and the completed, please send the completed checklist to Headquarters and close FMI per regional procedure.

Remove this FMI document from the customer site.

Appendix A

Labels

Figure 1 – Gantry Labels

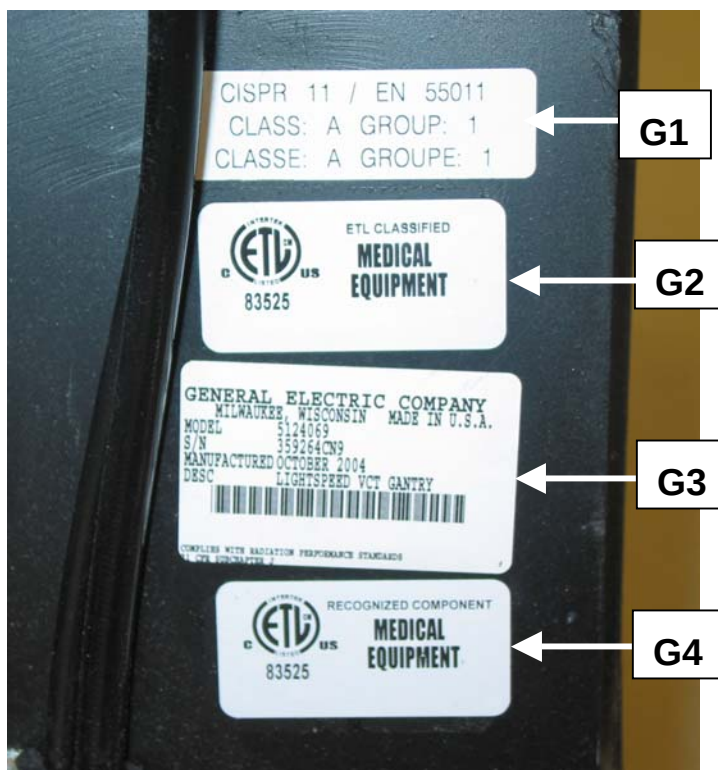


Figure 2 – Aux Box Labels



Appendix A – Continued

Figure 3 – Table Labels

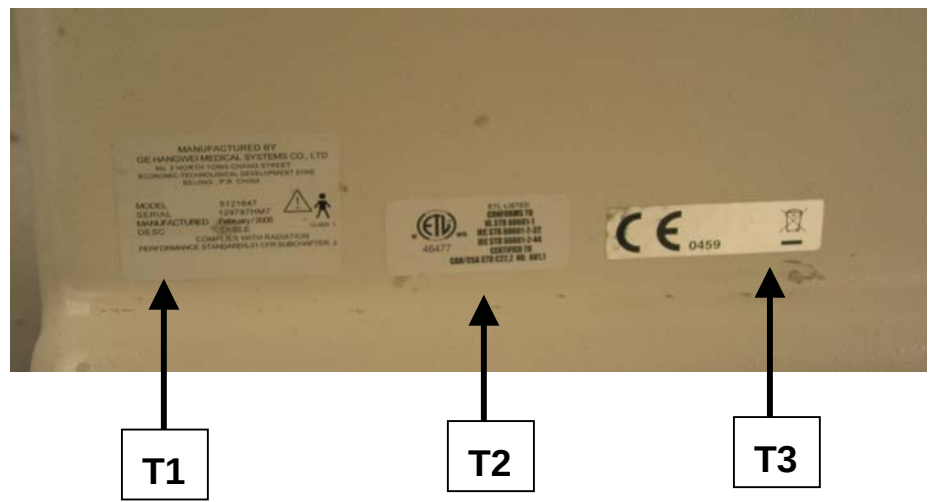
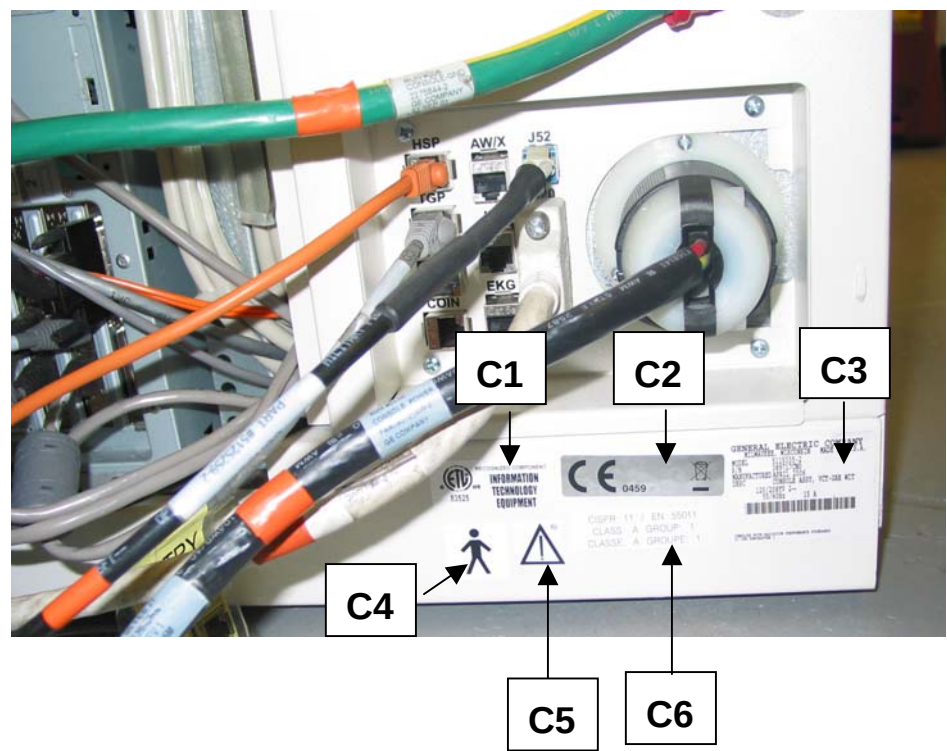


Figure 4 – Console Labels



Appendix A – Continued Labels

Figure 5 - PDU Labels1

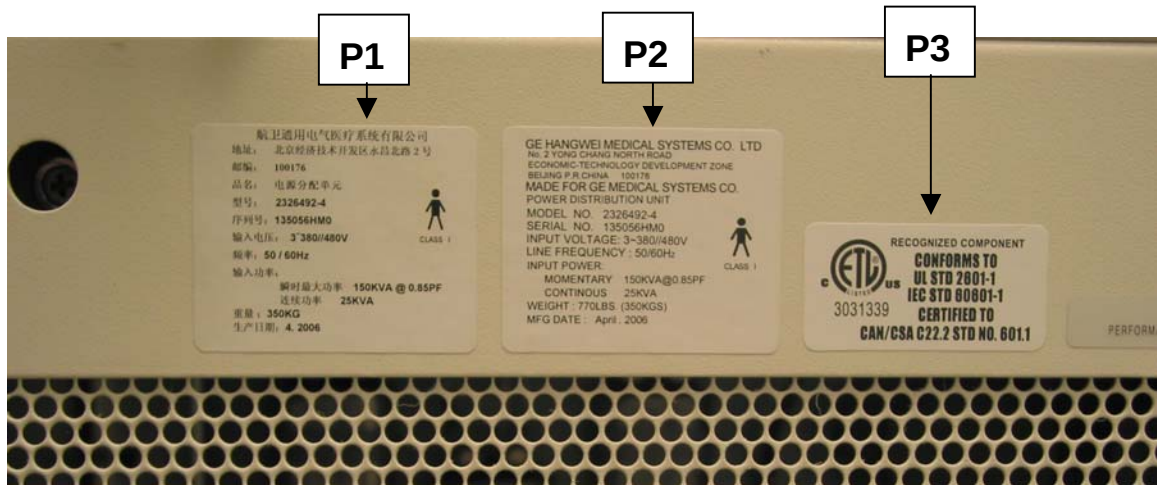
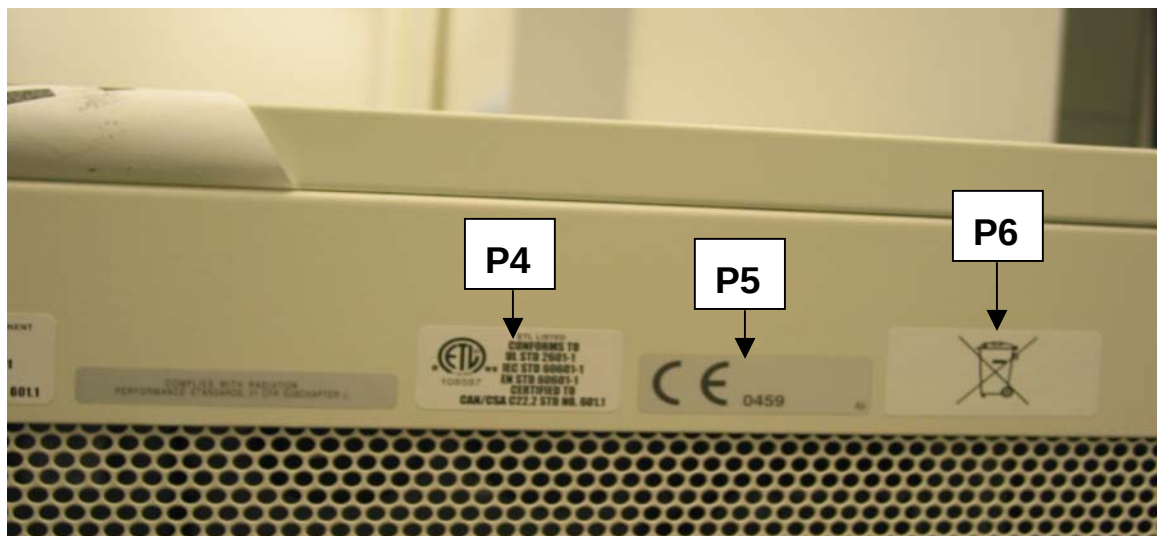


Figure 6 – PDU Labels2

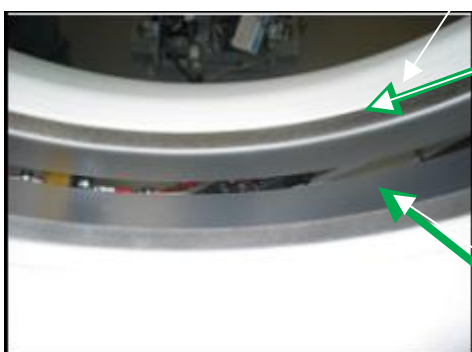


Appendix B

Installation of hardware Components

Please check and make sure that the following components are installed. If any are not installed, please contact Headquarters.

Gantry Covers



The correct part numbers are:

Front cover **2397075**

Rear cover **2397078**

If the system is not getting Mylar Window Errors during fastcal and the scan window doesn't slide back and forth, no action needed.



Scan window part number should be **2401577**. The correct part will have glazed area.

This cable (in the Gantry Top cover) should be a gray or black shielded harness, part number **5130600**. If it is black and red and not shielded, it is the old part number.

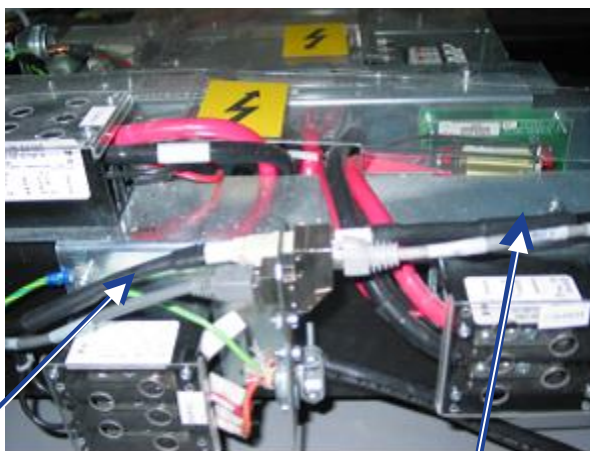
Appendix B – Contd.

Table Base

Global table front is shown in picture. There should be a **ferrite bead** on the Global table to gantry harness, **5129522**.



Gantry Base



Verify that **5125258** – fiber optics with black wrap is installed. Old version will be orange with no wrap. (HVDC power pan rear shown)

Verify that **5125259** – fiber optics with black wrap is installed. Old version will be orange with no wrap.

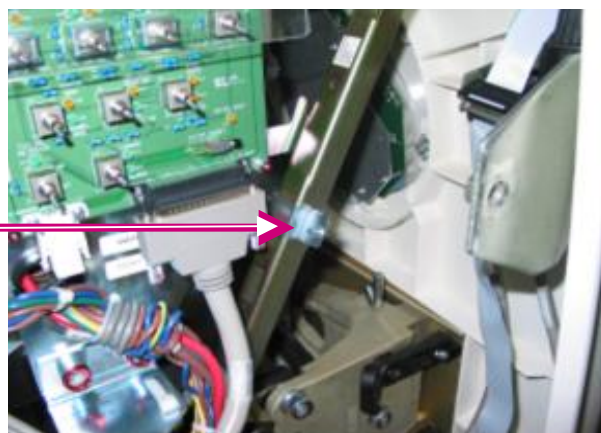
Appendix B – Contd.

Right side - Gantry

5126690 KEEPERS – right front
Verify these Keepers are
installed on gantry cover support
base

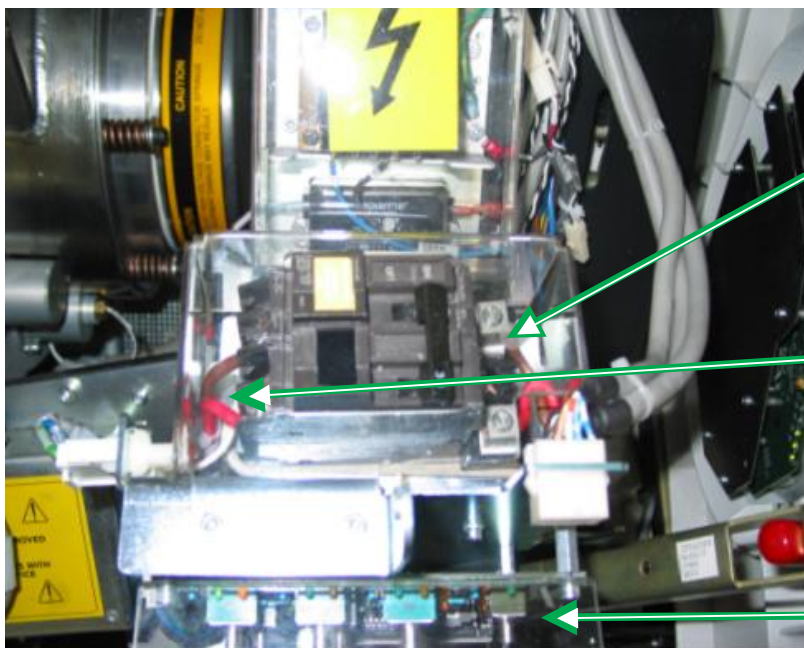


5126690 KEEPERS – right rear
Verify these Keepers are
installed on gantry cover support
base



Appendix B – Contd.

Right side – Gantry, looking down



Verify that harness, **5118424** (Filter to GFI harness) is installed.

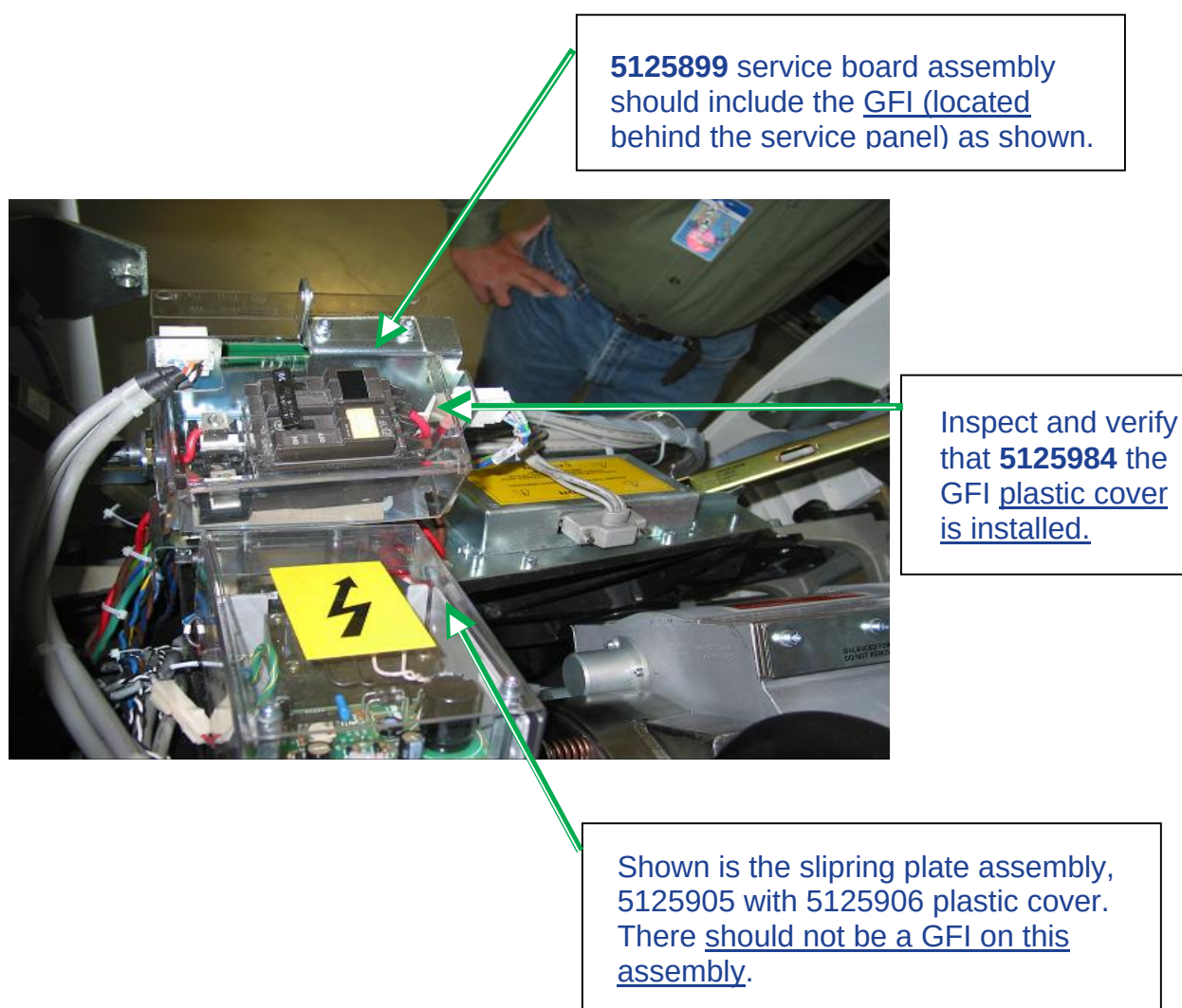
Verify that harness **5118296** (GFI to brush-block harness) is installed.

Service board assembly.

Appendix B – Contd.

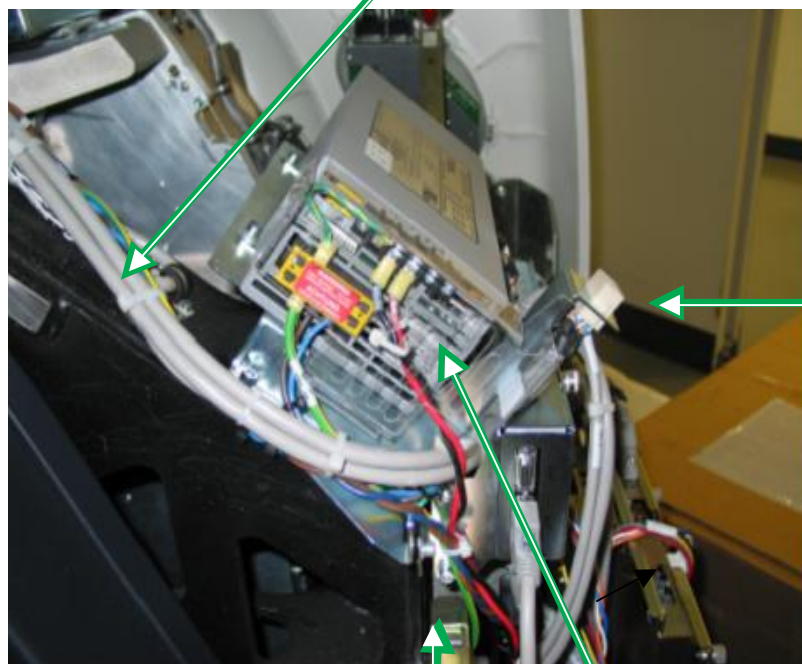
Picture from above right rear of gantry, looking out

GFI was moved from bottom (Slip Ring Plate Assembly) to the top (Service Switch Board Assembly). If moved, the plastic covers will be as shown below. If not, please contact headquarters.



Appendix B – Contd.

Left side - Gantry



Verify that **5130598** shielded gray or black harness is installed.

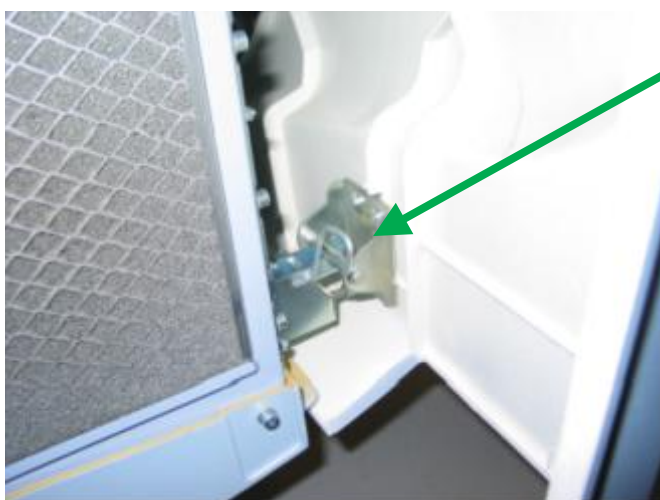
Verify **5128605** bulkhead bracket is installed and holds fan harness in place.

Verify that the Slam latch keeper hook is installed on the power supply assembly. Also verify that **5126690** slam latch keeper is on the slam latch.

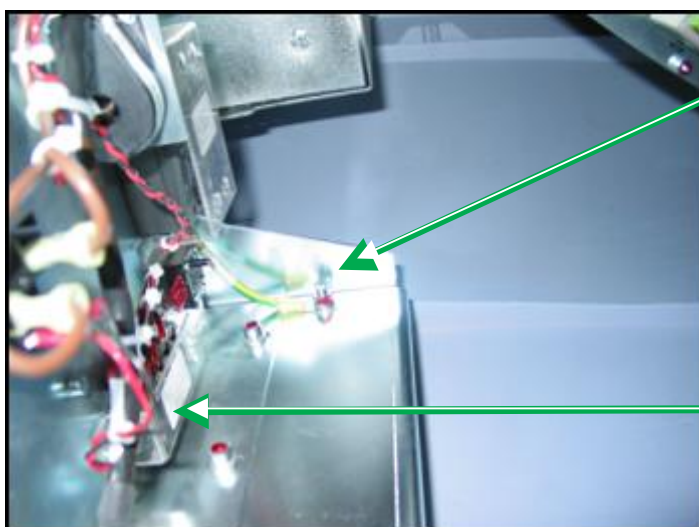
Verify **5128606** guard has air vents is installed.

Appendix B – Contd.

Left side - Gantry



Should have cover latch assembly **5122691** (lower left of gantry near blower box and filter as shown). Front cover shown attached.



There should be a ground wire attached to the base plate of the stationary side blower box (lower left of gantry).

Terminal strip should have a cover on blower box. Part number is **5119936**.

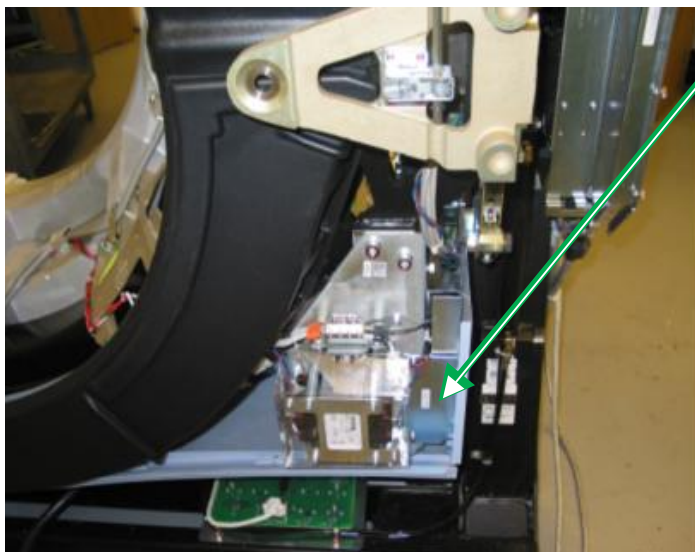
Appendix B – Contd.

Left side Gantry



Verify that **5124826** – left side panel is installed.

The side panel should be as shown.



Verify that **5124827** left side panel bracket is installed.

It holds the newer panel 5124826.

(Note: If the system has the side panel, the side panel bracket will be there.)

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